LECTURE – 8

QUICKSORT

Simple, efficient, in-place, 'stable'.

Simplified basic idea: Suppose the array is partitioned in two parts s.t.:

all the numbers in the first part $\leq$ all the numbers in the second part

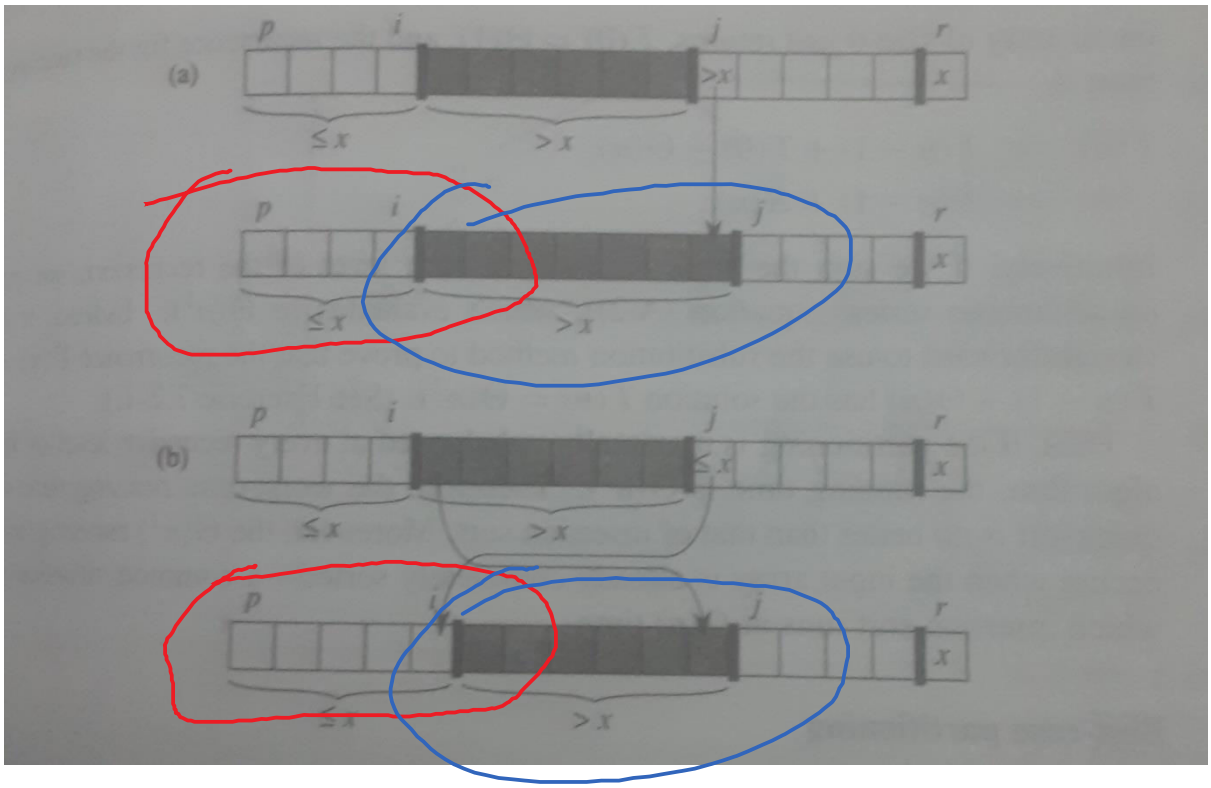
Then the two parts can be sorted independently of each other! **Recursively!**

```
QUICKSORT( A, p, r )
  if p < r                          // significance
     q = PARTITION( A, p, r );
     QUICKSORT( A, p, q-1 );
     QUICKSORT( A, q+1, r );        // q separates the
                                    // two parts
```

The main question is how to partition the array.

```
PARTITION( A, p, r )
  x = A[r];                 // pivot element
  i = p-1;
  //
  for j = p, r-1
    if A[j] ≤ x
      i = i+1;              // first part grows: A[j]
      EXCHANGE A[i], A[j]; // goes to first part
  //
  EXCHANGE A[i+1], A[r];    // move pivot element
                            // to its right place
  return i+1;               // and return its index
```

The figure below illustrates the main step in the loop.



Loop invariant:

At the start of each iteration of the **`for`** loop:

In the range given by $p \leq k \leq i$, $A[k] \leq x$ // smaller values

In the range given by $i+1 \leq k \leq j-1$, $A[k] > x$ // larger values

A[r] = x, which is the pivot element.
In the range: $j \leq k \leq r-1$, nothing is known yet about A[k].

Because of the single pass of the **`for`** loop, the running time of **`partition`** is easily seen to be $\Theta(n)$, where n = r-p+1. Worst case running time of quicksort is $\Theta(n^2)$. Best case and average case running time of quicksort is $\Theta(n \lg n)$.

Exercise: Array A[1..8] contains the numbers 2, 8, 7, 1, 3, 5, 6, 4. Show that the partition generated as above is 2, 1, 3, 4 || 7, 5, 6, 8 [CLRS Fig. 7.1].

Related points to note:

1. **Randomized** version of **`quicksort`**.
2. Meaning of **order statistics**.
3. Example of **sort algorithms with $\Theta(n)$ running time**.